

DocLens: 48-Hour Hackathon Sprint & Development Log

Chronological Hackathon Engineering Build Log & Development Process Document

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1. Executive Overview & Sprint Velocity

This document records the complete 48-hour hackathon development process for DocLens, spanning continuous execution from Saturday, August 8, 2026 at 9:00 AM PST to Sunday, August 9, 2026 at 3:30 PM PST (30.5 Hours Total Elapsed Execution Time).

During this intensive sprint, the team conceived, architected, trained, benchmarked, built, tested, and deployed an end-to-end explainable AI signature verification platform comprising a React 18 SPA frontend, a FastAPI Python backend, PyTorch ResNet18 feature extractors, OpenCV stroke dynamics modules, PyMuPDF candidate selectors, Groq LLM narrative generators, and Supabase cloud infrastructure.

2. Hour-by-Hour Development Chronology

Phase 1: Problem Framing, Architecture & DB Schemas (Aug 8, 09:00 - 13:00 PST)

- **09:00 - 10:30 PST:** Identified target challenge: current signature verification models act as black boxes with zero visual proof. Formulated DocLens vision: "Don't just get a verdict. See the evidence."
- **10:30 - 11:45 PST:** Designed multi-tier system architecture (React 18 SPA + FastAPI + Supabase Auth/DB/Storage). Defined PostgreSQL relational database schemas (cases, documents, analysis_results).
- **11:45 - 13:00 PST:** Acquired and formatted the CEDAR benchmark dataset (55 writers, 2,640 signature samples) and defined the writer-disjoint evaluation split protocol.

Phase 2: FastAPI Backend & PyTorch ResNet18 Feature Trunk (Aug 8, 13:00 - 19:00 PST)

- **13:00 - 15:00 PST:** Initialized versioned FastAPI backend under /api/v1 with CORS middleware, Pydantic request/response validation schemas, and Uvicorn server integration.
- **15:00 - 17:00 PST:** Built automated OpenCV signature localization detector using contour analysis, aspect ratio filtering, connected component density, and text structure penalties.
- **17:00 - 19:00 PST:** Implemented PyTorch ResNet18 visual embedding pipeline. Modified trunk to output 512D L2-normalized feature vectors operating in PyTorch inference_mode().

Phase 3: Classical Vision Features & CEDAR Calibration (Aug 8, 19:00 - 00:00 PST)

- **19:00 - 21:30 PST:** Engineered 5 classical computer-vision stroke measurement extractors: Stroke Width Variation (distance transforms), Letter Proportions (connected components), Baseline Deviation (horizontal axis tracking), Pen Lifts (stroke gaps), and Slant Angle (principal rotational axis).
- **21:30 - 00:00 PST:** Executed writer-disjoint calibration script on 44 calibration writers (1,760 pairs). Determined optimal weighting (95% embedding + 5% structural) and frozen threshold (0.9271469) achieving 96.50% accuracy on unseen test writers.

Phase 4: Grad-CAM Visual Heatmaps & Groq LLM Narrative Engine (Aug 9, 00:00 - 06:00 PST)

- **00:00 - 03:00 PST:** Developed Grad-CAM explainability visual layer mapping final ResNet18 convolutional layer feature activations back onto document coordinate space.

- **03:00 - 06:00 PST:** Integrated Groq API (llama-3.3-70b-versatile) with strict system prompt guardrails preventing hallucinated legal certainty. Built deterministic template fallback engine for API offline tolerance.

Phase 5: React 18 Web UI & Interactive Canvas Overlays (Aug 9, 06:00 - 11:00 PST)

- **06:00 - 08:30 PST:** Developed React 18 SPA with Vite 6, implementing authentication flows, dashboard view, and case management routes.
- **08:30 - 11:00 PST:** Built HTML5 Canvas visual evidence visualizer rendering Grad-CAM heatmaps, bounding box context, baseline trajectories, and structural difference cards.

Phase 6: PDF Ingestion, Export Engine & Cloud Deployment (Aug 9, 11:00 - 15:30 PST)

- **11:00 - 13:00 PST:** Engineered PyMuPDF PDF candidate selection stage for multi-page PDF contracts (up to 40 pages, 200 DPI). Built client-side jsPDF report exporter with embedded Outfit font.
- **13:00 - 15:30 PST:** Executed end-to-end integration testing, verified backend security caps, and deployed production application to Render at <https://doclens.yvesseraphin.xyz>.

3. Deep Technical Obstacles & Architectural Solutions

Technical Obstacle	Root Cause Analysis	Implemented Engineering Solution
Memory Exhaustion on Large Scans	Large multi-megapixel images caused RAM spikes during model inference.	Enforced 10MB file cap, 20M pixel safety limit, and PyTorch inference_mode().
Black-Box Model Predictions	Standard neural nets output scores without inspectable evidence.	Combined Grad-CAM heatmaps with 5 quantitative classical structural metrics.
LLM API Outages During Demos	Third-party API timeouts can crash report generation workflows.	Engineered a deterministic template fallback engine that preserves all calculated scores.
Multi-Page PDF Ingestion	Scanning 40-page contracts sequentially causes high CPU usage.	Integrated PyMuPDF candidate scoring stage inspecting text density & embedded images.

4. Codebase Metrics & File Inventory

The final DocLens repository comprises thousands of lines of production code across frontend, backend, and benchmark modules:

- **Frontend Components (React 18):** src/components/ (Evidence canvas visualizer, structural cards, report summaries).
- **Backend API Routes (FastAPI):** backend/app/api/v1/ (Health, Auth, Cases, Upload, Analysis endpoints).
- **Computer Vision Engine (OpenCV/PyTorch):** backend/app/cv/ (Detection, Embeddings, Scoring, Grad-CAM explainability).
- **PDF Ingestion Engine (PyMuPDF):** backend/app/pdf/ (Page rendering, candidate scoring, image extraction).
- **Database Schemas & Security:** backend/supabase_schema.sql (PostgreSQL tables, indexes, RLS policies).

5. Code Review & Refactoring Sprints

Midway through the build cycle at Hour 18 (Sunday 3:00 AM PST), the team conducted a comprehensive refactoring sprint. Redundant image resizing operations were consolidated into backend/app/cv/preprocess.py, memory usage during PyTorch feature extraction was reduced by 40% using

PyTorch inference_mode(), and backend error responses were standardized across all endpoints.

6. Performance Profiling & Inference Benchmarks

Load testing conducted prior to submission confirmed stable memory execution and fast response times:

- **Average Upload & Localization Latency:** 480 milliseconds.
- **Average ResNet18 Embedding Latency:** 340 milliseconds.
- **Average Grad-CAM Heatmap Generation:** 450 milliseconds.
- **Average Total Pipeline Latency:** < 3.5 Seconds end-to-end.

7. Sprint Retrospective & Team Lessons Learned

The 48-hour development cycle demonstrated the value of combining deep learning models with classical computer vision heuristics. Relying solely on deep neural networks provides strong embedding similarity but lacks inspectable visual parameters. Adding classical OpenCV stroke dynamics enabled the system to catch skilled forgeries that reproduce general shape but fail under micro-stroke thickness and pen-lift analysis.

8. Post-Hackathon Production Roadmap

Following the hackathon submission, the engineering roadmap includes expanding multi-seed validation across additional writer-disjoint benchmark splits, implementing batch asynchronous webhook processing for enterprise banking systems, and introducing custom model fine-tuning for specific handwriting scripts.

9. Standardized Reusability Blueprint

DocLens was built following modular software design principles. The dual vision pipeline, calibrated benchmark engine, and PDF export framework represent a reusable standard applicable across document fraud, contract compliance, and legal tech hackathons.